

Exploring engagement with music in everyday life using experience sampling methodology

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Abstracts

Recent qualitative research has highlighted differences in people's music-listening behaviour according to their level of involvement with and interest in music, yet these findings are mainly based on retrospective accounts of patterns of behaviour (Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009). Experience sampling methodology (ESM) is a valuable tool for studying music in everyday contexts, and it has been shown to increase people's conscious awareness of the role of music in their lives (Sloboda, O'Neill, & Ivaldi, 2001). Using ESM, the present study explored differences in people's everyday engagement with music by recruiting three different types of listener: those identified as having low, moderate, or high engagement with music. Over seven days, quantitative and qualitative data were collected from 25 young adults (aged 18–30) on what they were doing while hearing music (e.g., activities) and the functions/effects of music (e.g., reasons influencing choices). Post-study interviews with 23 participants then generated retrospective accounts about specific musical experiences. Analysis revealed two broad types of listener: the less engaged, who listened for fewer hours a week (min = 3 hours, mean = 12 hours), were less likely to be hearing self-chosen music, and were more likely to listen to music to pass time, out of habit or to help them feel less alone; and the highly engaged, who listened for a greater number of hours per week (mean = 21 hours, max = 40 hours), were more likely to be hearing self-chosen music, and were more likely to use music to evoke specific moods, create an atmosphere, or enhance an activity. The study confirmed the usefulness of ESM for investigating the complex (and interacting) factors involved in people's daily musical choices, and highlighted ways in which music can fulfil different functions concurrently. Findings show that future research on everyday musical behaviour should continue to account for the context of music listening in shaping responses to and uses of music; and account for individual differences in people's levels of engagement with music.

Keywords

engagement, everyday life, experience sampling methodology, functions, music

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Introduction

People in Western industrialized societies experience a great deal of music as they go about their daily lives (North, Hargreaves, & Hargreaves, 2004; Sloboda, O'Neill, & Ivaldi, 2001; Sloboda, Lamont, & Greasley, 2009). Self-reports from questionnaires (Juslin & Laukka, 2004) suggest that music is heard "several times a day" by 64% of respondents and only "once a day" by 18%. Experience-sampling studies indicate that music is being heard between 35% and 45% of the time by adults (Bailes, 2007; Juslin, Liljeström, Västfjäll, Barradas, & Silva, 2008; Sloboda et al., 2001), with nearly double this amount of exposure for young children (Lamont, 2008).

The opportunities for experiencing music are also more diverse than at any other point in history (cf. Hargreaves, Miell, & MacDonald, 2002). In everyday life, drawing on evidence from experience sampling, listeners have been found to experience music in a wide range of contexts and settings, from getting dressed in the morning at home through travelling to and from work to socializing in public places in the evenings (Juslin et al., 2008; North et al., 2004; Sloboda et al., 2001). While music is rarely the primary focus of people's attention as they go about their daily lives (Juslin & Laukka, 2004; Juslin et al., 2008; Sloboda et al., 2001; Sloboda, 1999), even in such "passive" listening conditions music is still noticed and continues to have effects. Juslin et al. (2008) noted mood change in 64% of listening episodes, and it was rare for people to state that music had no effect on them, even when they had no choice over the music (North et al., 2004; Sloboda et al., 2001).

The prevalence of music in everyday life can be explained by its capacity to fulfil a wide range of different functions (DeNora, 2000; Hargreaves & North, 1999; Sloboda, 1999). Evidence shows that adolescents engage in music listening to fulfil self-actualizing needs (e.g., to help get through difficult times), emotional needs (e.g., to relieve stress) and social needs (e.g., to be popular with others) (North, Hargreaves, & O'Neill, 2000; Tarrant, North, & Hargreaves, 2000; Saarikallio & Erkkilä, 2007). Older age groups (i.e., young adults through to older people) mainly listen for purposes of mood change and mood regulation, relaxation, reminiscence, catharsis, and company (DeNora, 2000; Juslin & Laukka, 2004; Lamont & Webb, 2010; Laukka, 2007; Sloboda, 1999; Sloboda & O'Neill, 2001). Finally, the most commonly cited reason for choosing to hear music across all ages is "enjoyment" (Lamont & Webb, 2010; North et al., 2000).

However, these results somewhat obscure the complexity of real-life listening situations. It has been argued that the functions and effects of listening to music need to be understood in terms of the listening context and in relation to the concurrent activities that might be taking place (DeNora, 1999; Hargreaves, North, & Tarrant, 2006; Juslin et al., 2008; Konečni, 1982). One way of capturing this is to use experience sampling methodology (ESM), which enables 'slices' of everyday life to be captured in a naturalistic setting and allows researchers to characterize the totality of listening experiences across a range of contexts and activities. ESM has been developed by Csikszentmihalyi & colleagues over the last 30 years (Csikszentmihalyi, 1975, 2000; Csikszentmihalyi & Lefevre, 1989; Hektner, Schmidt, & Csikszentmihalyi, 2007), and is broadly defined as "a research procedure for studying what people do, feel, and think during their daily lives" (Larson & Csikszentmihalyi, 1983, p. 41). As a psychological tool, Hektner et al. (2007) argue that ESM is invaluable for obtaining vast amounts of empirical, quantitative data, but also valuable as a qualitative research tool which allows an exploration of ways in which participants perceive their physical and social environments.

Studies that have used this method in relation to music listening have provided support for earlier questionnaire and self-report data showing that music is listened to in a range of situations and has a range of effects (Juslin et al., 2008; North et al., 2004; Sloboda et al., 2001). People are most likely to be listening to music while at home, while travelling, and in

shops and places of entertainment; their motivations for choosing music are highly context-dependent (e.g., highly arousing music while exercising); and music has been shown to serve a range of individual functions such as relaxation, gaining solace and creating or changing an atmosphere or mood. Social factors are found to influence the outcomes of music listening: people have a higher degree of liking for music when they are alone, and their reasons for choosing music vary as a function of who they are with. People's greater liking for music when they are alone is linked to the finding that they are more likely to have choice over music when alone, which relates to earlier work highlighting that the more choice people have over music the more likely it is to be associated with positive valued outcomes (North et al., 2004; Sloboda et al., 2001). Experience sampling also appears to raise listeners' conscious awareness of their everyday music use (Sloboda et al., 2001).

Despite a growing body of evidence on the functions of music in everyday life, there has been less focus on music's ability to fulfil different functions concurrently or on how functions might relate to effects. North et al. (2004) asked participants to report reasons for choosing music when they had choice over music, and effects of the music only when they had no choice. A more pertinent line of enquiry would be to ask whether music does fulfil the functions for which it has been chosen, which necessitates gathering data on functions and effects of music irrespective of whether the music is chosen or not. Sloboda et al. (2001) began to unravel some of these complexities by asking their participants to report patterns of mood change. They found that listeners can sometimes experience an increase in one mood state while simultaneously experiencing a decrease in another. For example, listeners reported experiencing increases in positivity (e.g., relaxing) together with decreases in arousal and present-mindedness (e.g., disengaging with surroundings), which suggests that music can have more than one concurrent effect. This chimes with evidence from self-report diary studies indicating that music is often felt to serve multiple functions (Lamont & Webb, 2010; van Goethem & Sloboda, *in press*), although these studies similarly do not relate reasons for choosing to hear music to the effects the music had.

A further limitation of the existing research on everyday experiences of music to date is that it has treated its participants as a homogeneous sample. Studies have either explored young adult populations (e.g., Lamont & Webb, 2010; North et al., 2004; Sloboda et al., 2001) or targeted other specific demographic groups such as older people (e.g., Hays & Minichiello, 2005; Laukka, 2007). Although Juslin et al. (2008) asked participants to report on their musical education and experience, no study has yet compared the behaviours of individuals with differing levels of musical interest or experience. However, recent qualitative research has shown that people respond to and use music in very different ways (Caldwell, MacDonald, Duncan, & Thompson, 2009; Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009; Lonie, West, & Wilson, 2008). For example, Greasley and Lamont (2006) found differences between more and less engaged music listeners (defined by self-ratings of the importance of music in participants' lives, the extent to which they enjoyed music listening, and the amount of music owned). Highly engaged participants listened to as much music as possible, preferred to listen to self-chosen music (whenever possible), were involved in a greater number of music-related activities (e.g., performing, attending concerts/gigs), had a more detailed knowledge of different styles of music, and higher conscious awareness of the ways in which they engaged with music and its effects on their mood. Conversely, those who were less engaged were less concerned with listening to music regularly, listened to the radio rather than specific styles, and their engagement with music was strongly influenced by others (i.e., they were more likely to acquire music as a result of other people, and were more likely to be hearing music as a result of others' musical choices). Lonie (2009) found similar results in relation to emotional self-regulation and awareness of music as a resource for health and wellbeing. These findings

highlight the need to differentiate between participants in a more specific and structured manner and to explore whether such differences in conscious awareness might also be found in more fine-grained investigations of everyday life music-listening experiences.

The current study

Experience sampling has been shown to be a useful tool for capturing daily musical experiences in the real-life contexts in which they naturally occur (e.g., Juslin et al., 2008; North et al., 2004; Sloboda et al., 2001) and raising listeners' awareness of their everyday music use, thus enabling a more accurate picture of musical behaviour to be obtained (Sloboda et al., 2001). Within this, however, two issues have been identified as ripe for further investigation: the nature of the sample and the notion of intended functions and effects.

First, as illustrated above, qualitative evidence (e.g. Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009) highlights differences in people's listening behaviour according to their level of involvement with and interest in music. However, this is based on retrospective accounts of patterns of engagement with music, and it is not clear to what extent these map onto everyday music listening. The current study uses ESM to explore differences in engagement types and whether they map onto differences in real-time listening. Since previous research suggests that more engaged listeners are more consciously aware of how they use music in everyday life (Greasley & Lamont, 2006), ESM may help uncover musical behaviours that individuals with a low level of involvement with music are less aware of.

Second, very few studies have explored the ways in which music may fulfil different functions concurrently. Do people give broadly consistent reasons for choosing to hear music, or is there a spread of reasons according to situation and context (as previous literature would suggest)? Furthermore, how is this related to level of engagement? Do some individuals 'get more' out of music, and note that music is fulfilling a number of different functions simultaneously, while others report a single reason and effect of choosing/hearing music? The second aim of the study is to provide insights into these questions by exploring both reasons for choosing music and the effects the music subsequently has.

In summary, the current ESM study will explore the role of music in people's everyday lives, investigating aspects of behaviour which have been mapped in previous studies (e.g., where people are when hearing music, what they are doing, and so on), but with a novel focus on differences in engagement types and the functions/effects of music. Based on existing research it is predicted that listeners will vary in their amount of music listening and choice over music, with more engaged participants showing higher levels of both; that more engaged participants should predominantly use music for purposes of mood-regulation and to help them carry out activities, while less engaged participants should mainly use music to pass the time, keep them company or for reminiscence; and that less engaged participants will be hearing music more frequently as a result of others. While this is speculative, it is also expected that more engaged participants will report that music fulfils more functions and effects than will less engaged participants, given that they are more likely to conceptualize music as a positive health and wellbeing resource in everyday life.

Method

The current study followed the fundamental principles of experience sampling methodology (Csikszentmihalyi & Lefevre, 1989; Hektner et al., 2007). The response form combined

contextual information on what participants were doing while hearing music with factors relating to the functions/effects of music listening. Post-study interviews then generated further information about specific music episodes in the form of retrospective accounts which were combined with the response booklets to explore participants' thoughts and feelings at the time of hearing music.

A pilot study was carried out to check the validity of the ESM response form and procedure. While some minor changes were made to the response form as a result, the pilot showed the method was feasible, participants understood what was being asked of them, and a wide range of musical experiences could be captured using the selected response frequency during the week.

Participants

The study consisted of 25 volunteer participants – 21 women, 4 men, age range 18–29 years old (mean age = 20.4 years; $SD = 2.96$) – selected from a larger sample who had filled out a profile questionnaire designed to identify individuals with differing levels of engagement with music (as defined by Greasley & Lamont, 2006, based on the emerging factors of levels of preferred control over music listening, quantity of music owned, organization of music, and knowledge about and motivations for musical purchases/downloads). This short profile questionnaire enabled three groups of participants to be defined. Highly engaged participants preferred to have control over the music they were hearing, owned a great deal of music, organized their music according to a system, and so on. Moderately engaged participants owned less music and tended to have less control and organization, while less engaged participants tended to listen to other people's music, owned very few CDs, and so on. Based on responses to this questionnaire, the final sample consisted of 12 'more', 7 'moderately', and 6 'less' engaged participants.

Participants were drawn from the undergraduate psychology population at a university in the West Midlands of England. The gender imbalance was apparent in both the screening and the final sample, and reflected the generally higher numbers of women enrolled on psychology courses in the UK. However it may also have resulted from a gender bias in volunteering, as previous research found that women were more likely to volunteer for ESM studies and more likely to respond to a greater number of signals than men (Hektner et al., 2007). All participants are referred to here by pseudonyms to preserve their anonymity.

Materials: The response form

The response form (Appendix A) combined forced-choice questions, rating scales and open-ended questions covering different elements of musical behaviour. Forced-choice categorical responses to questions (e.g., location: home, shops, etc.) were included so that the form could be completed quickly (as in North et al., 2004). However, previous findings on the variations in people's responses to music and musical preferences (Greasley & Lamont, 2006) highlighted the need for open-ended questions to explore participants' thoughts and feelings at the time of hearing music in more detail (as in Sloboda et al., 2001).

The first contextual section asked participants to identify where they were, what they were doing, and their reasons for engaging in the activity. For music episodes only, participants were also asked who they were with, the mode of delivery, and the technology (if any) used in the music episode. They were then asked to describe the music they were hearing using an open-ended response format, given the problems raised earlier with closed categorical responses (cf. Greasley, Lamont, & Sloboda, under review).

The third section on choice covered the degree of choice participants had over the music (rated on an 11-point scale); their reasons for choosing the music, and/or the effects it had on them, using a tick box with predetermined reasons drawn from earlier research; any other reasons that influenced their choice; and whether they had considered other music. Participants were asked to report on both the reasons for choosing to hear music, if applicable, and the effects of hearing music, to explore the relationship between predicted and actual effects of music (cf. Greasley & Lamont, 2006). In addition, an open-ended response format encouraged participants to reflect on any other reasons influencing their musical choices.

A final section contained two open-ended questions asking participants to note any additional information about what was happening and how they were feeling when hearing music.

Post-study interviews

Post-study interviews were carried out with 23 of the 25 participants (unfortunately two participants were not available). The interview schedule was designed to investigate experiences of participating in the study and to obtain retrospective accounts of specific music episodes during the week (based on Sloboda et al., 2001). Participants were asked whether the week had been typical in terms of daily activities and the amount of music they were exposed to, and whether they had been more consciously aware of music than usual. They were also asked whether they had consciously changed their music habits or the ways in which they thought about music as a result of participating in the study, as one way of exploring the potential problem of reactivity (Hektner et al., 2007).

Response booklets were read before the interviews and participants were asked to elaborate on five music episodes that had occurred during the week. Unlike the random selection of music episodes for elaboration employed by Sloboda et al. (2001), episodes were purposefully chosen to represent a range of situations (e.g., alone/with others; music chosen/non-chosen; range of contexts). To ensure the reliability and validity of the data, transcriptions and analysis were undertaken with respect to Lincoln and Guba's (1985) criteria of transferability and dependability which included providing rich, detailed descriptions of participants' experiences, and keeping detailed records of each stage of the research process (e.g., interview transcripts, details of participant recruitment).

Procedure

Participants volunteered to take part, and were given further information on the study and asked to provide contact details and an indication of their normal waking hours on each day of the week (to ensure that the study was as non-intrusive as possible). All participants who volunteered owned a mobile telephone (as expected, given that levels of mobile telephone ownership for UK residents in the 16–24 and 25–34 age groups were 96% and 95% respectively; Hoare, 2007).

Text messages were sent to participants' telephones from a single number, which acted as an immediate channel through which participants could contact the researcher. Participants were sent a text message five times a day, for seven consecutive days (a total of 35); these were spread out randomly across participants' waking hours to capture the whole day. Participants differed greatly in the hours they kept; some gave identical hours for all seven days while others gave different hours for each day based on work commitments.

Participants were asked to have their response booklet (and a pen) with them at all times, and to fill out a form as soon as they received a text. If this was not possible (e.g., if they were driving a car), they were instructed to fill in the response form at the next convenient moment. If the participant was not hearing music at the time they responded, and had not heard music in the last three hours, they were instructed to complete the first section only. If participants were hearing music, or had heard music in the previous three hours, they were instructed to complete all sections of the form.

Participants were encouraged to write on the response form if the categories provided did not cover what was actually happening, and instructions emphasized that there were no correct or wrong answers and that answers were confidential. At the end of the week, participants were asked to return the response booklets as soon as possible, and were invited to take part in a post-study interview. After completion of the post-study interviews, all participants were debriefed and thanked for their time. The study was approved by the appropriate School of Psychology ethics committee in advance.

Results

This section combines closed and open-ended responses generated by the response forms with retrospective accounts collected in the post-study interviews. In the following, an *episode* represents a time at which a participant was sent a text and responded, while a *music episode* is a time when a participant responded and was hearing, or had been hearing music. The length of a music episode was defined by the participants themselves (e.g., a music episode could be 5 minutes or 2 hours). Analyses were performed on data at the response level (i.e., by episode), participant level (i.e., looking across all episodes for each participant) and group level (i.e., comparing the three engagement groups). Analysing data at the response level is problematic in relation to significance tests because the data points are not independent (Juslin et al., 2008), thus group comparison data are reported here at the participant level.

Compliance rates, responses delays and the prevalence of music

In total, 872 text messages were sent to the 25 participants (with 3 instances of researcher error). From these, 866 response forms were completed, with an average compliance rate of 99.2%. In post-study interviews the majority of participants reported they had enjoyed taking part and found it interesting, which may explain this very high level of compliance. Of the responses, 541 (63.1%) were made within 10 minutes of receiving the text message (as the form took several minutes to complete, this was considered a fairly immediate response); 100 (11.6%) responses were made within 30 minutes, and a further 87 (10.1%) were made within one hour. In contrast, only 33 (3.8%) of the responses involved long delays (more than four hours). There were discernible individual differences in the speed with which participants responded to the texts (the most efficient participant responded immediately in 91.4% of episodes, while the least efficient responded immediately in only 24.2% of episodes). These differences did not appear to be related to age, gender or level of engagement. Overall, given that shorter delays indicate higher reliability, the finding that 75% of the responses were made within 30 minutes of receiving the message was indicative of the motivation and commitment participants showed throughout the study.

Out of 866 episodes, music was being heard at the time of text in 302 (35%) episodes, and had been heard in the last three hours in a further 152 (18%) episodes. In total, music was

being heard or had been heard since the previous signal in over half of the total number of episodes. For reliability purposes, a music episode was only included in the analysis if there was less than two hours between the time at which participants were hearing music and the time they filled out the form. Two key findings informed this decision. First, participants were hearing music on average for 59 minutes per episode, with some (9 out of 25) averaging more than 70 minutes per episode, and they were hearing music for two hours or more in 19% of music episodes (85 out of 454). Second, 93.1% of responses in music episodes were made within two hours of hearing music. Valid music episodes were defined as those where the music ended less than two hours before the response was completed, or if the total duration of listening was greater than the difference between the music ending and the response. Using these criteria, 407 music episodes were retained for the main analysis (90% of the total number of music episodes).

Amount of music heard

On average, participants were hearing music for an hour in each episode, and for approximately 18 hours a week (see Table 1). The maximum total amount of time a participant was hearing music for was 39 hours and 30 minutes (one more engaged participant); the minimum was 2 hours and 40 minutes (one moderately engaged participant). Nearly half of the participants heard music for more than 20 hours during the week, and of these, four participants were hearing between 30 and 40 hours. Table 1 outlines the amount of time spent hearing music for participants collectively, and for each engagement group separately.

Although the more engaged group were hearing music for longer during episodes than the moderately or less engaged groups, there were no significant differences between these listening durations ($F(2, 22) = 1.881, p = .176$). Similarly, although the more engaged group heard more music across the week than the moderately or less engaged groups, this difference was also not significant ($F(2, 22) = 1.191, p = .323$). This is likely to be due to the distribution of values, as can be seen from the standard deviations, which highlighted considerable individual variation within, as well as between each engagement group.

Technology

Listening to recorded music was the dominant mode through which participants were experiencing music. Participants were hearing recorded music in 291 episodes (75%); hearing music through broadcast (e.g., TV, radio) in 80 episodes (21%); and hearing music live in only 17 episodes (4%). There were differences between the engagement groups. The 17 live music

Table 1. Time spent hearing music: General statistics and group comparisons

Level of engagement group	Average no. of music episodes	% of episodes in which hearing music	Average length of time hearing music per episode in minutes (SD)	Average total time spent hearing music during the week in hours and minutes (SD)	
More	20	58.5	69 (26)	21:10	(10:00)
Moderate	15	43.3	48 (26)	16:24	(9:51)
Less	17	49.5	51 (20)	14:51	(4:51)
All participants	18	52.4	59 (26)	18:19	(09:07)

episodes fell into three categories and were all experienced by more engaged participants. Eleven episodes involved participants hearing DJs mixing live in a nightclub or on the radio (on vinyl/CD decks) or were instances in which they themselves were mixing on decks (this only applied to the two DJs in the sample); 4 episodes were in church (e.g., piano music at a wedding, hymns in mass); and 2 episodes involved participants watching friends playing instruments (e.g., listening to housemates playing their guitars).

The most frequently used technology in the music episodes was participants' computers, which accounted for 153 of the 407 music episodes (39%). This reflects a trend amongst students, who tend to copy music to their computers and leave CDs at home when returning to university, or to purchase music as downloads more generally. Use of a stereo system (either at home or work) accounted for 72 music episodes (18%), with the third most frequent type of technology being radio, which was used in 47 music episodes (12%). Music was being heard via a PA system in 35 (9%) episodes; car stereo in 30 (8%) episodes; and portable music player in 22 (6%) episodes. Participants heard music via TV (e.g., MTV) in 16 music episodes and through their mobile phone in two episodes, which points towards the increasing use of different media as channels through which music is heard. For example, Vikki (F: 29yrs) wrote: "Kerrang TV is the popular mode of music delivery in our home" on one of her response forms. There was only one instance in which a participant was hearing music as a result of attending a concert.

Contexts

Overall, while an approximately equal number of music and non-music episodes occurred when participants were at home, at someone else's house, and in social settings (e.g., shops, gym), as expected from previous findings, participants were more likely to hear music when travelling (75% of travelling episodes) and in a place of entertainment (88%), and less likely to hear music in the workplace (including educational settings: 30%) (see Table 2). There were differences by engagement group. While participants across all three groups were most frequently hearing music at home, the less and moderately engaged groups were more likely to be

Table 2. Frequency (%) of contexts in which music episodes and non-music episodes took place

	No. of episodes (% of music, non-music and total episodes respectively)					
	Music episodes		Non-music episodes		Total episodes	
Home	259	(64.1)	264	(65.0)	523	(64.6)
Workplace (incl. educational setting)	25	(6.2)	59	(14.5)	84	(10.4)
Transportation (i.e., car, bus, train)	38	(9.4)	13	(3.2)	51	(6.3)
Social/pub	22	(5.5)	18	(4.5)	40	(4.9)
Gym/exercise	3	(0.7)	2	(0.5)	5	(0.6)
Shops	5	(1.2)	8	(2.0)	13	(1.6)
Entertainment (i.e., bar, nightclub)	22	(5.5)	3	(0.7)	25	(3.1)
Someone else's house	30	(7.4)	39	(9.6)	69	(8.5)
Total	404		406		810	

hearing music at someone else’s house (9%, 14%) than the more engaged group (4%), who were more likely to be hearing music in the workplace (10%) than the less and moderately engaged groups (0%, 2%). This provides some support for the notion that those in the less engaged group would more frequently be hearing music as a result of others.

Activities

The activities participants were engaged in during episodes were coded in line with the categorization of activities in Sloboda et al. (2001). Overall, while an approximately equal number of music and non-music episodes occurred when participants were engaged in personal maintenance (e.g., washing, getting dressed, cooking: 18% and 19% respectively) and self-directed work (e.g., computing, reading for study, sorting finances/bills: 19%, 17%), as expected from previous findings, participants were more likely to be hearing music while travelling (10% of music episodes, 4% of non-music episodes) and while engaging in active leisure pursuits (e.g., socializing, eating out, exercising: 31%, 16%), and less likely to be hearing music while engaged in passive leisure activities (e.g., watching TV, sunbathing: 7% of music episodes, 25% of non-music episodes) and during personal being activities (e.g., sleeping: 3%, 12%) (see Table 3).

Participants varied greatly in the types of activities they were engaging in when hearing music, and these differences did not appear to be related to engagement group. For example, Jed (M: 19yrs, more engaged) was predominantly engaged in leisure activities such as socializing and playing computer games (59%) when hearing music, whereas Leanne (F: 19yrs), another more engaged participant, was mainly engaged in self-directed work such as writing and revising (71%). Across all groups, there were individual differences in participants’ use of music while working. Some reported they did not work with music on. For example, Anna (F: 18yrs, more engaged) emphasized that “when I revise, I have to have silence”. Conversely, others

Table 3. Frequency (%) of main activities during music, non-music and total episodes

	No. of episodes (% of music, non-music and total episodes respectively)					
	Music episodes		Non-music episodes		Total episodes	
Time fillers	3	(0.7)	5	(1.2)	8	(0.9)
Personal						
Being	11	(2.7)	51	(12.4)	62	(7.6)
Maintenance	77	(19.1)	74	(18.1)	151	(18.6)
Travel	40	(9.9)	15	(3.7)	55	(6.8)
Leisure						
Music	19	(4.7)	0	(0)	19	(2.3)
Passive	27	(6.7)	104	(25.4)	131	(16.1)
Active	127	(31.4)	68	(16.6)	195	(24.0)
Work						
Self	80	(19.8)	71	(17.3)	151	(18.6)
Other	19	(4.7)	21	(5.1)	40	(4.9)
Other	1	(0.3)	1	(0.2)	2	(0.2)
Total	404		410		814	

reported that they could not work without music on. For example, Lucy (F: 20yrs, less engaged) stated that she always worked with music, and explained why:

“if it’s complete quiet I’ll get distracted and like, wander off into daydreaming or start walking round the room, but if I’ve got music there, then it kind of, it kind of distracts me from my work, but kind of helps me concentrate by making me sit there and do the work, ‘cause there’s something else on in the background.”

Overall, participants’ responses to (and retrospective accounts of) music episodes in which they were engaged in self-directed work suggest that music is an essential tool in facilitating mental concentration and focus. These examples also highlighted the use of specific styles of music to accompany specific activities. For example, Melissa (F: 19yrs, more engaged) reported that she would only ever revise to a specific album (*Native Spirit* by Dan Gibson) and no other music.

Participants reported listening to music attentively as a sole activity in only 2.3% of all episodes, and there was a difference between engagement groups. Of the 23 episodes in which participants had been listening to music as a main activity, 19 were reported by the more engaged group, with only two each reported by the less and moderately engaged groups.

In the post-study interviews, participants were asked how important the music was to the activity they were engaged in at the time. In some episodes, music was reported to be central. For example, Naomi (F: 26yrs, more engaged) was at home exercising on her stepper for an hour during one episode. She had chosen various rock/metal tracks to help her exercise, create a mood, change an existing mood, and for enjoyment. In turn, she reported the music had these effects on her, noted that she exercised daily, and had chosen the music to motivate her and get her adrenaline pumping. In her interview, Naomi was asked to elaborate on this episode, and describe some of the characteristics of music she felt ‘helped’ while engaged in this activity. She explained that her regular painkillers made her tired, the music had helped her to feel energetic, and that the lively and rhythmic beat had assisted with stamina levels, taken her mind off exhaustion and kept her motivated. In other episodes, the importance of music was fundamentally determined by contextual factors. For example, in one episode, Anna (F: 18yrs, more engaged) was at work (working on a checkout in a superstore) and had reported that the music helped her carry out duties and pass the time. In her interview, when asked how important music was to the activity in hand, she stated that it was more salient at certain times than others: “it’s nice to listen to something in the background, but when we’re busy you don’t tend to notice ‘cause I have to focus on other things”. Thus music was not always integral to activities, even when participants had reported that the music had fulfilled a number of different functions.

The role of others

Overall the highest frequency of music episodes (54.9%) occurred when participants were alone; however, there were considerable individual differences across all participants. For example, Naomi (F: 26yrs) experienced 20 music episodes, and was with her partner or family members in 72% of these, and alone in far fewer episodes (28%). In contrast, Ally (F: 19yrs) experienced 19 music episodes and was alone in 16 of these (84%), and with people she knew in only 3 (16%). Results show that musical choices are often ‘negotiated’ with others present in the listening context, as exemplified by two episodes in which Naomi was in the car with her partner travelling to and from a weekend away (to attend a wedding). On the journey there, she reported that she had no choice over the music, and had to “suffer” her partner’s music for 4 hours. The only effect of music she reported was that her partner liked it. On the journey back, she reported

that they had considered his music (Elton John), and then her music (Evanescence), and reached a compromise by listening to neither (Bryan Adams). On this occasion she said that the music had a number of positive effects. There were also a number of episodes in which participants had purposefully chosen music because they thought the person(s) they were with would like it. Furthermore, participants reported they would have left certain situations because they disliked the music if they had not been with others at the time. In one episode, Sarah (F: 21yrs) explained difficulties in conversing with a friend in a bar because they were being exposed to '80s electro music at a loud volume: "If I hadn't have been with her, I'd have walked out". Thus the presence/absence of others plays a fundamental role in determining people's responses to music.

Musical style

Participants were hearing a wide variety of musical styles during the course of the study, including popular music (e.g., rock, indie), classical music (e.g., opera, orchestral), world music (e.g., African, Vietnamese) and electronic dance music (e.g., techno, drum and bass). A full list of musical styles, as written on the response forms by the participants themselves, is in Appendix C. In order of highest frequency, the ten most frequently reported styles where style was explicitly reported are shown in Table 4.

A key finding is that participants were listening to more than one style in 25% of episodes (19% to 2 styles, 5% to 3, and 1% to more than 3). There were three main patterns of response. In some cases, the music they were hearing was described as a cross-over in musical styles (e.g., a fusion of rock and pop, commonly notated as rock/pop). In others, the music being heard was a mixture of styles (e.g., chart music on the radio which covered pop, rock, indie). In yet other cases, participants had purposefully chosen to listen to different styles within the same listening episode (e.g., rock and indie). While there was a tendency for more engaged participants to report a greater variety of musical styles, data show that all participants were listening to a range of different musical styles during the course of the week. This underlines the importance of employing an open-ended response to explore the music people were hearing, rather than a limited number of tick-box categories.

Level of choice over music

Overall, participants were mainly listening to self-chosen music (see Figure 1). Data were categorized into three levels of choice for purposes of analysis (low choice 0–3, moderate choice

Table 4. Ten most frequently reported styles

Style	Episodes	Percentage
Pop	46	19.2%
Soundtrack	22	9.2%
Rock	15	6.3%
Indie	13	5.4%
Drum 'n' bass	13	5.4%
RnB	12	5%
Classical	10	4.2%
Dance	8	3.3%
Pop/Rock	7	2.9%
Rock/Indie	7	2.9%

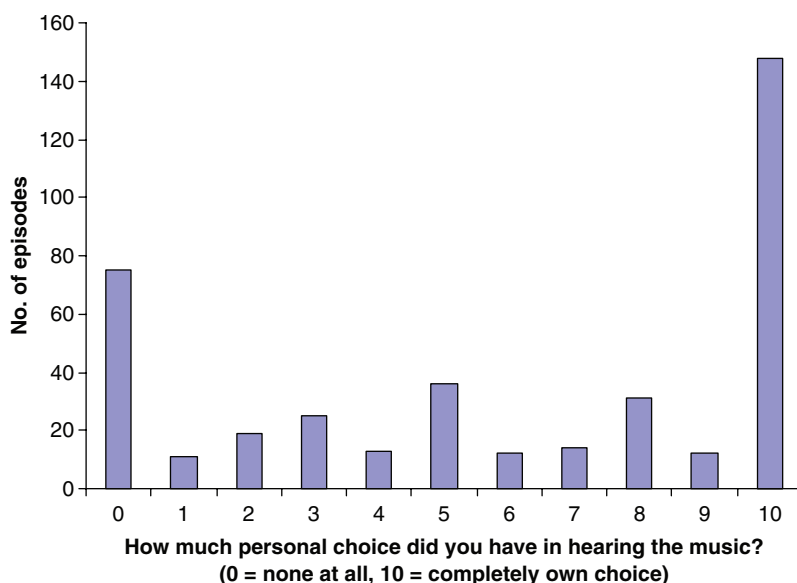


Figure 1. Level of choice participants had over music

4–7, high choice 8–10). In line with previous research, participants were more likely to have choice over music at home (59% of home music episodes) and in the workplace (64%), and less likely to have choice over music when in public places (6%); and more likely to have high choice over music when they were alone (66%), and low choice when they were with others (24%).

When participants had a high level of choice, they were more likely to report choosing music for enjoyment (78% of total music episodes) and relaxation (68%), and more likely to report music had helped them to concentrate/think (66%). There were more instances of music having little or no effect and music having annoyed participants in low choice episodes compared to episodes in which they had moderate or high choice. These patterns suggest a strong positive link between level of choice over hearing music and valued outcomes, thus supporting the findings of earlier work that participants would experience more beneficial effects from hearing self-chosen music. There were differences by engagement group. More engaged participants had a greater level of choice over music (mean = 6.88, $SD = 1.71$) than moderately (mean = 4.64, $SD = 1.93$) and less engaged (mean = 4.83, $SD = 2.84$) participants, and these differences were marginally significant ($F(2, 22) = 3.34, p = .054$).

Functions/effects of music

Analyses of the functions/effects of music listening were performed across the dataset, by engagement group, and individually.

A total of 818 reasons for choosing to hear music and 1134 effects of hearing music were reported in the music episodes across the week. As predicted, there was a high frequency of music episodes in which participants had chosen to hear music for enjoyment (this was chosen in 62% of all music episodes), for relaxation (42%), to help carry out activities (29%), pass the time (28%) and create an atmosphere (23%). However, music was less frequently chosen to create, accentuate or change specific moods than anticipated (17%, 7% and 5% respectively). Also

as expected, frequently cited effects of music listening were enjoyment (66% of music episodes), relaxation (36%), because it passed the time (29%) and helped create an atmosphere (23%), and there was a low frequency of episodes in which music had little or no effect (8%) or had been annoying/irritating (7%).

A key finding was that participants had chosen to hear music for three or more reasons in 58.2% of episodes (average of three per episode) and thus were predominantly choosing music to fulfil a number of different functions at the same time.

The extent to which reasons for choosing music mapped onto the effects of music was explored. Response forms were analysed individually, highlighting many episodes in which music had the intended effects, and many episodes in which the effects had not matched the reasons for choosing music. For example, in one episode, Alice (F: 19yrs) was engaged in self-directed work and had chosen music to help her concentrate/think, to bring back memories, because she liked listening to it, and to listen to the lyrics. Rather than helping her to concentrate or bringing back memories, she reported that the music actually distracted her and created a particular mood. In another episode, Chloe (F: 19yrs) had chosen music to bring back memories, for enjoyment and because her partner liked it. The music had these effects, namely, it brought back memories, and she and her partner enjoyed listening to it. However, the music also changed an existing mood, created a different (happy) mood and helped her to relax.

Comparison of engagement groups showed differences in the number and types of functions/effects reported. While less engaged participants reported between 1 and 5 reasons for choosing music (mean = 2.42, *SD* = .92) and moderately engaged between 1 and 6 (mean = 2.45, *SD* = 1.10), some more engaged participants reported up to 10 or 12 reasons for choosing to hear music per episode (mean = 3.75, *SD* = 1.65). These differences were marginally significant ($F(2, 21) = 2.69, p = .091$) and provide some support for the notion that more engaged participants would choose to hear music for a greater number of reasons. In relation to effects, although there were no significant differences ($F(2, 22) = .490, p = .619$), while less engaged participants reported between 1 and 5 effects of hearing music (mean = 2.65, *SD* = .69) and the moderately engaged between 1 and 6 (mean = 2.73, *SD* = 1.14), some more engaged participants reported up to 10 or 11 effects of hearing music (mean = 3.21, *SD* = 1.59). To

I chose the music . . .		The music . . .	
☒	to help me concentrate/think	☒	helped me concentrate/think
☐	to distract me	☐	distracted me
☐	to help me relax	☒	helped me relax
☒	to help me carry out activity	☒	helped me carry out activity I was doing
☒	to help me create right atmosphere	☐	helped me create right atmosphere
☐	to bring back certain memories	☒	brought back certain memories
☒	to create an emotion/mood	☒	created an emotion/mood
☐	to accentuate an emotion/mood	☒	accentuated an emotion/mood
☐	to change an existing emotion/mood	☐	changed an existing emotion/mood
☒	because I really like listening to it	☒	I liked listening to it
☐	thought person(s) I was with would like	☐	the person(s) I was with liked it
☐	to help me pass the time	☒	helped me pass the time
☐	to help me feel less alone	☒	helped me feel less alone
☐	to listen to the lyrics	☐	had little or no effect
☒	out of habit	☐	annoyed/irritated me

Figure 2. An example of reported functions/effects of music in one listening episode

illustrate this, Figure 2 provides an example of one more engaged participant's reasons/effects of hearing music. Melissa (F: 19yrs) was at home revising in one episode, and reported she had chosen the music for six different reasons and experienced 10 different effects.

There were also differences in the types of reasons/effects according to engagement groups. Less engaged participants were choosing to hear music out of habit, to pass time and to feel less alone more frequently than moderately and more engaged participants, as expected. However, contrary to prediction, the less engaged group did not use music to bring back memories more frequently than the other two groups (see Table 5).

More engaged participants chose music to create the right atmosphere and to listen to lyrics more frequently than moderately and less engaged participants. However, moderately engaged participants were more frequently choosing music to accentuate a mood, create a mood, and help them concentrate/think. Thus the prediction that more engaged participants would be listening to music most frequently for mood regulation was not statistically supported. The qualitative accounts suggested that more engaged participants were using music to fulfil therapeutic needs more frequently than the other groups. For example, Naomi (F: 26yrs) was using music for mood regulation in the majority of her music episodes, regardless of the activity she was carrying out (e.g., exercising, working, driving). In one episode she was driving in the car and had chosen to hear Evanescence – *Fallen* to relax, to help her drive, to create a mood, for enjoyment, out of habit, and to listen to lyrics. In turn, she reported that the music had helped to relax her, helped her carry out the activity she was engaged in, created the 'right' mood, and that she had enjoyed listening to it. She wrote on the response form:

Table 5. Frequency of episodes in which the following statements were reported as a reason for choosing to hear music by level of engagement group (less, moderately, more)

I chose the music . . .	Total	No. of episodes (% of total reasons per engagement group)					
		Less engaged		Moderately engaged		More engaged	
to help me concentrate/think	55 (6.7)	7 (5.9)	13 (9.9)	35 (6.2)			
to distract me	30 (3.7)	3 (2.5)	8 (6.1)	19 (3.3)			
to help me relax	113 (13.8)	19 (16.1)	16 (12.2)	78 (13.7)			
to help me carry out/enhance the activity I was doing	77 (9.4)	3 (2.5)	19 (14.5)	55 (9.7)			
to help me create the right atmosphere	61 (7.5)	5 (4.3)	4 (3.1)	52 (9.1)			
to bring back certain memories	22 (2.7)	2 (1.7)	4 (3.1)	16 (2.8)			
to create an emotion/mood	34 (4.2)	5 (4.3)	4 (3.1)	25 (4.4)			
to accentuate an emotion/mood	18 (2.2)	0 (0)	5 (3.8)	13 (2.3)			
to change an existing emotion/mood	14 (1.7)	0 (0)	0 (0)	14 (2.5)			
because I really like listening to it	167 (20.4)	28 (23.7)	25 (19.1)	114 (20.0)			
because I thought the person(s) I was with would like it	39 (4.8)	7 (5.9)	6 (4.6)	26 (4.6)			
to help me pass the time	74 (9.0)	15 (12.7)	14 (10.6)	45 (7.9)			
to help me feel less alone	26 (3.2)	6 (5.1)	3 (2.3)	17 (3.0)			
to listen to the lyrics	37 (4.5)	2 (1.7)	3 (2.3)	32 (5.6)			
out of habit	51 (6.2)	16 (13.6)	7 (5.3)	28 (4.9)			
Total	818	118	131	569			

"I listen to it daily because it's my favourite . . . the songs help me to find a sense of identity, and help me understand the self-loathing I have on the inside. It is also a great tool for releasing any tension in a non-physical and non-aggressive way."

When asked in her post-study interview how important the lyrics were in allowing her to find a sense of identity within the music, she replied: "the lyrics in Evanescence songs are very original, deeper and more honest than any I've heard before, they say things that I feel and daren't express to anyone else". Her descriptions of how the music made her feel on the response form (i.e. at the time she was listening) corresponded with her retrospective account; the songs on this particular album were a catalyst for venting negative emotions.

Other accounts provided by more engaged participants contained similar assertions of their awareness of the ways in which they used music in daily life. For example, when asked whether she had been more consciously aware of music during the week, Vicki (F: 29yrs) stated: "No, I'm always aware of what I do with music, it's part of who I am". Steve (M: 25yrs) clearly stated which reasons he did (and did not) select and why:

"I was aware that I didn't go for the memories one a lot at all, I felt that there was a distinct lack of that, and I didn't do 'helped me feel less alone', 'helped pass the time', er, listen to the lyrics, I didn't do 'listen to the lyrics' 'cause I don't generally listen to music with lyrics a lot, and 'out of habit', well, maybe I should have put that more, but that's kind of just a general personality thing, rather than a, you know, a habitual thing."

Further group differences were found in relation to the effects of hearing music (see Table 6). As predicted, less engaged participants reported that music had helped them to pass time and made them feel less alone more frequently than moderately and more engaged participants. Conversely, more engaged participants reported that music had changed a specific mood and created the 'right' atmosphere more frequently than moderately and less engaged participants, who were more likely to report that the person(s) they were with liked the music as an effect of hearing music. Less engaged participants also reported that music had annoyed them more frequently than moderately and more engaged participants. For example, Adam (M: 23yrs) was at a friend's flat having a cup of coffee and chatting. They were trying to watch a holiday video his friend had made, but all they could hear was the neighbour's music. On the response form, Adam ticked the effect 'it annoyed/irritated me' and wrote: "the annoying girl upstairs (above my mate's flat) was playing her music *really* loud". However, overall, annoyance was one of the least common effects of music. Participants rarely experienced negative effects of hearing music, regardless of their level of engagement.

Only one participant in the sample did not appear to derive any functions/effects from music listening. Rebecca (F: 18yrs, less engaged) had reported on the profile questionnaire that she usually listened to music other people were playing and had never purchased an album before. She was recruited because she appeared to have a very low level of engagement with music and this was borne out in her experiences of music in the study. Even though Rebecca stated that she had been more consciously aware of music during the week, she emphasized that she did not, in her own words, "get more from music" as a result. Of the 10 music episodes she experienced, music 'had little or no effect' in five of these. In three of these five episodes, she stated that the person she was with (her boyfriend) enjoyed listening. This supports the proposition that less engaged listeners are more likely to be hearing music as a result of other people, rather than choosing music personally.

Table 6. Frequency of episodes in which the following statements were reported as an effect of hearing music by level of engagement group (less, moderately, and more)

The music . . .	No. of episodes (% of total effects per engagement group)							
	Total		Less engaged		Moderately engaged		More engaged	
helped me concentrate/ think	62	(5.5)	13	(6.0)	16	(6.7)	33	(4.9)
distracted me	58	(5.1)	9	(4.1)	15	(6.3)	34	(5.0)
helped me relax	137	(12.1)	32	(14.8)	22	(9.2)	83	(12.3)
helped me carry out/ enhance the activity I was doing	99	(8.7)	10	(4.6)	25	(10.4)	64	(9.5)
helped me create the right atmosphere	88	(7.7)	11	(5.1)	11	(4.6)	66	(9.7)
brought back certain memories	45	(4.0)	6	(2.8)	13	(5.4)	26	(3.8)
created an emotion/mood	49	(4.3)	4	(1.8)	14	(5.8)	31	(4.6)
accentuated an emotion/ mood	37	(3.3)	4	(1.8)	4	(1.7)	29	(4.3)
changed an existing emotion/mood	18	(1.6)	0	(0)	2	(0.8)	16	(2.4)
I liked listening to it	254	(22.4)	55	(25.4)	52	(21.7)	147	(21.7)
The person(s) I was with liked it	85	(7.5)	19	(8.8)	20	(8.3)	46	(6.8)
helped me pass the time	110	(9.7)	30	(13.8)	25	(10.4)	55	(8.1)
helped me feel less alone	39	(3.4)	12	(5.5)	7	(2.9)	20	(2.9)
had little or no effect	28	(2.5)	5	(2.3)	8	(3.3)	15	(2.2)
annoyed/irritated me	25	(2.2)	7	(3.2)	6	(2.5)	12	(1.8)
Total	1134		217		240		677	

Finally, there were no patterns in reasons/effects chosen according to the time of day or day of the week (although overall participants were more likely to be experiencing music live at the weekend). There may be two explanations for this: first, there were considerable individual differences between participants in the reasons/effects they chose; second, the sample consisted mainly of students who have days which are likely to be less structured.

Reactivity

Participants were asked whether they had changed their music habits as a result of taking part in the study. Analysis highlighted three broad patterns. Some reported they had experienced no change in awareness or behaviour as a result of taking part; some reported there had been an increase in awareness but no change in behaviour; and others reported there had been an increase in awareness and changes in their behaviour, but that these changes in behaviour had been minimal. For example, Lucy (F: 20yrs) and Sarah (F: 21yrs) tried to listen to less music, but both reported that it “hadn’t worked” and that they had still experienced as much music as they normally would (if anything, slightly more). Steve (M: 26yrs) purposefully chose to hear certain music on a couple of occasions, but reported that he was behaving in a typical manner

95% of the time. These instances of reactivity thus did not appear to pose any significant threat to the overall validity of the results.

Looking across engagement groups, data from more engaged participants suggested that it only confirmed what they already knew about their engagement (i.e., in terms of their uses of music), while data from less engaged participants was mixed. For some less engaged participants, the study served to highlight functions of music in their lives, while others reported that while they had been more consciously aware of music, they did not feel it had affected the way they thought about or listened to music.

Discussion

The present results confirm a range of existing findings regarding the prevalence, use and effects of music in everyday life. These include: considerable amounts of music listening, yet low incidence of listening to music as a focused activity; a range of contexts and activities in which people frequently hear music; individual differences in the extent to which people are hearing music in the presence of others; a predominance of listening to self-chosen music, and more choice over music to accompany personal maintenance and active leisure pursuits, at home or when travelling, and alone; and finally that people are mainly listening to music for enjoyment and to relax (cf. Juslin & Laukka, 2004; Juslin et al., 2008; Lamont & Webb, 2010; Laukka, 2007; North et al., 2004; Sloboda, 1999; Sloboda et al., 2001). Furthermore, people are listening to a wide range of styles, which underlines the need to map the music being heard using open-ended methods (cf. Greasley, Lamont, & Sloboda, under review). The use of pre-determined tick-box categories to explore music in everyday life (e.g., North et al., 2004) is highly unlikely to take into account all the styles people are listening to.

A major contribution made by the current study is the findings relating to people's levels of engagement in everyday life. As was predicted from previous qualitative research (e.g., Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009), the current results confirm that more highly engaged listeners are more likely to listen to a greater amount of music in any given week, and are more likely to be listening to self-chosen music. More engaged listeners are also more likely to be listening to music in the workplace than less and moderately engaged listeners, who are more likely to be hearing music at someone else's house. This provides support for the notion that less engaged individuals would more frequently hear music as a result of others (cf. Greasley & Lamont, 2006).

There are also differences in the number and type of functions/effects that music fulfils. Those who are more highly engaged choose to hear music for a greater number of reasons and are more likely to experience a greater number of effects of music. They are more likely to use music to create the right atmosphere or help carry out/enhance activities and to report that music changes a specific mood than less engaged listeners, who are more likely to choose music out of habit or to pass the time and report that music makes them feel less alone and that the person(s) they are with like(s) it. Furthermore, the more highly engaged are more likely to be listening to music attentively and experiencing music in live contexts.

In addition to these differences in behaviour, there are differences in levels of awareness of music. More highly engaged listeners show greater awareness of the ways in which they engage with music and the effects of music on their mood, while less engaged listeners have lower levels of awareness. The present study provides further evidence that participation in ESM studies increases awareness over the course of a week (cf. Sloboda et al., 2001).

While occasionally subtle differences emerge between all three groups, and sometimes two groups cluster together (the most commonly occurring pattern in the data being similarities

between less and moderately engaged groups, with differences between these groups and the more engaged group), results suggest two broad categories of listener type: the less and the more engaged, with these two groups showing different patterns of music use, choice, awareness and effect.

Another major contribution of the present study is in highlighting the extent to which people choose to listen to music to fulfil different functions simultaneously (music being chosen for an average of three reasons per episode). While listening to music typically fulfils these goals, in many cases additional (unanticipated) effects are experienced. Thus even when people have low choice over hearing music, it is generally shown to have positive effects on them. These results support the finding of Juslin et al. (2008) that people predominantly experienced positive effects of hearing music, and have implications for health psychology: listening to music is one way in which people can actively improve their psychological wellbeing and health (cf. Batt-Rawden & DeNora, 2005).

Methodological considerations

Overall, participants were highly committed to the study, and the current findings illustrate rapid and compliant responses to daily texts. However, individual differences were found in the speed with which participants responded to text messages (cf. Sloboda et al., 2001), and also differences in the amount they wrote on the response forms (i.e., how much information they provided). ESM researchers have suggested that personality factors are likely to affect the types of ways in which people respond (Bolger, Davis, & Rafaeli, 2003; Reis & Gable, 2000), but music psychology research has yet to explore this fully.

The current findings show participants experience an average of three music episodes per day, which suggests that the five daily signals used here provide sufficient data points to accurately reflect daily music-listening experiences. However, as illustrated by the open-ended responses, sampling over a week (as many existing ESM studies do) cannot account for participants' previous experiences, which are likely to influence and shape what they listen to at any given time. Future research could use ESM to explore people's engagement with music over longer time periods (e.g., months) in order to map the ways in which musical choices are informed by recent experiences with preferred music. Given that the current research and previous research (e.g., Sloboda et al., 2001) has shown that most people who take part in ESM studies exploring musical behaviour are committed to the task, they might be willing to participate for longer periods. A second direction could be the exploration of people's engagement with a new piece of music (or album, style etc.) that they themselves have recently purchased and brought to the researcher's attention. This could highlight ways in which the contexts of other music (i.e. existing preferences and patterns of music use) influence the manner in which people engage with new/different music, and would undoubtedly provide further evidence to suggest that people listen to a wide range of styles during a typical week.

Another consideration is that the characteristics of engagement as reported here are based on young adults' musical experiences, and these may (or may not) map onto different populations in the same way. If the current ESM study was conducted with professional musicians such as performers or conductors (i.e., those who engage with music on a daily basis as part of their jobs), would it be meaningful for them? Would they use music in similar ways when they were not engaging in live rehearsal/performance? Future research needs to explore differences between representative samples of participants. This could include mapping differences between those with a high level of musical training and those with no training, and individuals in a variety of vocational music jobs (e.g. professional musician, music lecturer, sound engineer and so on).

Finally, there were considerable individual differences between highly engaged listeners (e.g., Melissa) and less engaged listeners (e.g., Rebecca) in the current study. Melissa reported that she could not go a day without listening to music, while Rebecca reported that she derived little if any functions/effects from music listening. Future studies might also distinguish between participants on the basis of personality types, as while ESM takes the social context of listening behaviour into account, it does not account for intra-individual factors which may affect the ways in which people respond to and use music.

Conclusion

The present study has gone beyond existing research using forced-choice responses (e.g., North et al., 2004) by combining forced-choice and open-ended responses. This has resulted in the collection of rich contextual data in each music episode, further strengthened by participants' retrospective accounts in post-study interviews. In addition, by encouraging participants to write freely on the form, it has been possible to explore their thoughts and feelings even further. The current study has thus exemplified an in-depth and open-ended approach to the exploration of music's role in people's everyday lives using both quantitative and qualitative research methods. It has consolidated previous findings in relation to uses of music in everyday contexts and furthered our understanding of similarities and differences in the musical behaviours of people with differing levels of engagement with music.

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Appendix A

ESM Study Response Form

Date: _____ Time received text: ____ am/pm Time Filled Out: ____ am/pm

At the time you received the text message, were you hearing any music?

☐ YES ☐ NO If YES, please fill out this form with respect to the time you were texted

If NO, have you heard any music in the last three hours?

☐ YES ☐ NO If YES, please fill out this form with respect to the time you last heard music

Please also make a note of the time at which you were hearing the music: ____ am/pm

If NO, please just fill in Section A with respect to the time you were texted

SECTION A

Where were you?

- ☐ Home ☐ Workplace (incl. educational setting) ☐ Transportation (i.e. car, bus, train)
☐ Social/Pub ☐ Gym/Exercise ☐ Shops ☐ Entertainment (i.e. bar, nightclub)
☐ Someone else's house

What was the MAIN thing you were doing? (please go into as much detail as possible)

- ☐ What were the principal reasons why you were doing this particular activity? (tick as many as apply)
☐ It was a basic part of my routine (e.g. something that is so habitual that I don't really think about my motivations for doing it)
☐ I had to do it (i.e. it was part of my work, duty, responsibility)
☐ I wanted to do it (i.e. it gave me pleasure, satisfaction, fulfillment)
☐ I had nothing else to do (i.e. I was filling in time)
☐ Other (please state): _____
-

SECTION B

Who were you with when you were hearing music? (tick as many as are applicable)

- ☐ Alone ☐ Partner ☐ Person/people you live with ☐ Family member(s) ☐ Friend(s)
☐ Stranger(s) ☐ Acquaintance(s) ☐ Person/people you work with ☐ Professional(s)
 (e.g. dentist)

What music were you hearing? (please fill in as much as you can – genre, album, artist/band, performer, song etc.)

What was the mode of delivery?

- ☐ Live music ☐ Recorded music ☐ Broadcast (i.e. radio or TV)

What technology was involved in the music listening episode?

- ☐ Radio ☐ TV ☐ Phone ☐ Walkman (CD, MD & MP3 player, iPod) ☐ Computer
☐ Car stereo (radio, tape, CD) ☐ Stereo system (at home or at work) ☐ Vinyl/CD decks
☐ PA System (public place) ☐ Busker(s) ☐ Concert ☐ Other (please state)

How long did you hear the music for? _____

SECTION C

How much personal choice did you have in hearing the music (please circle)?

None at all 0...1...2...3...4...5...6...7...8...9...10... Completely own choice

If you chose the music, or had *some* choice over it, fill in both columns. If you didn't choose the music, just fill in the right hand side. Tick each box that applies.

I chose the music . . .	The music . . .
☐ to help me concentrate/think	☐ helped me concentrate/think
☐ to distract me	☐ distracted me
☐ to help me relax	☐ helped me relax
☐ to help me carry out/enhance the activity I was doing	☐ helped me carry out/enhance the activity I was doing
☐ to help me create the right atmosphere	☐ helped me create the right atmosphere
☐ to bring back certain memories	☐ brought back certain memories
☐ to create an emotion/mood	☐ created an emotion/mood
☐ to accentuate an emotion/mood	☐ accentuated an emotion/mood
☐ to change an existing emotion/mood	☐ changed an existing emotion/mood
☐ because I really like listening to it	☐ I liked listening to it
☐ because I thought the person(s) I was with would like it	☐ the person(s) I was with liked it
☐ to help me pass the time	☐ helped me pass the time
☐ to help me feel less alone	☐ helped me feel less alone
☐ to listen to the lyrics	☐ had little or no effect
☐ out of habit	☐ annoyed/irritated me

Are there any other reasons that influenced your choice? Please explain.

Did you consider other music before you settled on this particular piece of music? (if so, what did you consider and why didn't you put it on?)

SECTION D

Please write below any additional information or comments about what was happening and/or how you were feeling when you were hearing music.

Appendix B

List of all musical styles 25 participants were hearing

African	Jazz
Asian	French jazz
Blues	Love songs
Chill out	Latin
Chinese	Metal
Classical	Musicals
Opera	Muzak
Orchestral	Pop
Orchestral band	Chart
Piano music	Cheese
Country	Classics
Dance	Oldies
Electronica	Punk
French discotech	Rap
Europop	Encore rap
Trance	Reggae
Drum and bass	Rock
Jump-up dnb	Acoustic rock
Dark dnb	Soft rock
Emo	Punk rock
Folk	Emo rock
Funk	RnB
Garage	Soundtrack (e.g. film/TV theme)
Old skool garage	Swing
Gospel	Specific time periods (e.g. 60s, 80s, 90s, now)
Goth	Ska
Heavy metal	Soul
Hip-hop	Trip-hop
House	Techno
Funky house	Housey techno
Tech house	Minimal techno
Industrial	Vietnamese
Indie	Worship
US Indie	

Explorando el compromiso con la música en la vida cotidiana utilizando la experiencia metodológica de muestreo

La investigación cualitativa reciente ha puesto de relieve diferencias en el comportamiento de la gente que escucha música según su nivel de compromiso y de interés con la música; sin embargo, estas conclusiones se basan principalmente en los recuentos retrospectivos de modelos de comportamiento (Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009). La experiencia metodológica de muestreo (ESM) es una herramienta valiosa para estudiar la música en situaciones cotidianas, y se ha demostrado que aumenta la conciencia de la gente consciente del papel de la música en sus vidas (Sloboda, O'Neill, & Ivaldi, 2001). Utilizando la ESM, el presente estudio explora las diferencias en la participación cotidiana de la gente con la música mediante el reclutamiento de tres tipos diferentes de oyente: los identificados como de bajo, moderado o alto compromiso con la música. Durante más de siete días se recogieron datos cuantitativos y cualitativos de 25 adultos jóvenes (de 18 a 30 años) sobre lo que ellos estaban haciendo mientras escuchaban música (por ejemplo, actividades) y las funciones y efectos de la música (por ejemplo, las razones que influyen en las opciones). Las entrevistas post-estudio con 23 participantes generaron cuentas retrospectivas sobre las experiencias musicales específicas. El análisis reveló dos grandes tipos de oyentes: los menos comprometidos, que escucharon menos horas de música a la semana (mínimo: 3 horas, media: 12 horas), con menor probabilidad de escuchar música elegida por ellos mismos, y más propensos a escuchar música para pasar el tiempo, por costumbre o para sentirse menos solos; y los altamente comprometidos, que escucharon un gran número de horas por semana (media: 21 horas, máximo: 40 horas), que tenían más probabilidad de estar escuchando música auto-elegida, y eran más propensos a utilizar música para evocar estados de ánimo específicos, crear una atmósfera o realizar una actividad. El estudio confirma la utilidad de ESM para investigar los complejos (e interactivos) factores que intervienen en las opciones musicales cotidianas, y pone de relieve las vías en que la música puede cumplir distintas funciones simultáneamente. Los resultados muestran que las futuras investigaciones sobre el comportamiento musical cotidiano deben continuar para tener en cuenta el contexto del oyente musical en la formación de respuestas a y a los empleos de la música; y tener en cuenta las diferencias individuales en los niveles de compromiso de la gente con la música.

Esplorazione del coinvolgimento musicale nella vita quotidiana attraverso la metodologia del campionamento dell'esperienza

La ricerca qualitativa recente ha evidenziato differenze nel comportamento relativamente all'ascolto musicale delle persone a seconda del loro livello di coinvolgimento con interesse per la musica. Queste scoperte si basano principalmente su precedenti descrizioni di modelli comportamentali (Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009). La Metodologia del Campionamento dell'Esperienza (ESM) è un valido strumento per lo studio della musica nei contesti della quotidianità, e ha dimostrato di aumentare la consapevolezza delle persone sul ruolo che la musica ha nella loro vita (Sloboda, O'Neill, & Ivaldi, 2001). Attraverso l'ESM questo studio ha esaminato le differenze nel coinvolgimento musicale quotidiano delle persone. Sono stati formati tre gruppi di ascoltatori suddivisi in base al tipo di coinvolgimento musicale: basso, moderato alto. Nell'arco di sette giorni sono stati raccolti dati quantitativi e qualitativi su 25 giovani adulti (di età compresa tra 18 e 30 anni) su quello che svolgevano durante l'ascolto della musica e le funzioni/effetti della musica (per esempio, ragioni che influenzano scelte). Le

interviste condotte successivamente con 23 partecipanti hanno portato a risultati sulle specifiche esperienze musicali. L'analisi ha rilevato due grosse tipologie di ascoltatore: quello meno coinvolto, che ascolta musica per poche ore alla settimana (con un minimo di 3 e una media di 12 ore) i quali ascoltavano musica prevalentemente non scelta da loro e come semplice passatempo, al di fuori delle loro abitudini e per sentirsi meno soli; e quelli fortemente coinvolti, che ascoltavano musica per un maggior numero di ore (una media di 21 e un massimo di 40 ore), i quali ascoltavano musica principalmente scelta da loro e che utilizzavano la musica per evocare precisi stati d'animo, creare una determinata atmosfera o migliorare un'attività. Lo studio ha dimostrato l'utilità dell'ESM nella ricerca dei fattori complessi (e che interagiscono tra loro) coinvolti nelle scelte musicali quotidiane delle persone, e ha messo in evidenza i diversi modi in cui la musica può esercitare contemporaneamente varie funzioni. I risultati spingono alla prosecuzione della ricerca futura sui comportamenti musicali quotidiani per acquisire dati e informazioni nel contesto dell'ascolto musicale che diano delle risposte anche sull'utilizzo della musica; e dati sulle differenze in ciascun individuo nei diversi livelli di coinvolgimento musicali.

Une exploration de l'engagement musical dans la musique à l'aide de la méthode d'échantillonnage d'expériences

Une récente recherche qualitative a mis en évidence des différences entre les personnes dans les comportements d'écoute musicale selon leur niveau d'implication et leur intérêt pour la musique, encore que ces résultats soient basés principalement sur des comptes-rendus rétrospectifs de types comportementaux (Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009). La méthode d'échantillonnage d'expériences (*experience sampling methodology*, ESM) est un outil précieux pour l'étude de la musique dans les contextes de tous les jours, dont on a pu montrer qu'il augmentait la conscience qu'ont les individus du rôle de la musique dans leur vie (Sloboda, O'Neill, & Ivaldi, 2001). En utilisant l'ESM, la présente étude a examiné les différences dans l'engagement quotidien qu'ont les individus avec la musique, en recrutant trois types différents d'auditeurs selon leur niveau - bas, moyen, élevé, d'engagement musical. Pendant sept jours, on a recueilli des données quantitatives et qualitatives auprès de 25 jeunes adultes (âgés de 18 à 30 ans) sur ce qu'ils font (quelles activités) quand ils écoutent de la musique et les fonctions/effets de la musique (par ex., quelles raisons influencent les choix). Les interviews post-étude de 23 participants ont ensuite fourni des comptes rendus rétrospectifs à propos d'expériences musicales spécifiques. L'analyse a révélé deux grands types d'auditeurs: d'une part, les moins engagés, qui écoutaient quelques heures par semaines (minimum 3 heures, en moyenne 12 heures), étaient les moins susceptibles d'écouter de la musique qu'ils auraient eux-mêmes choisie, et plus susceptibles d'écouter de la musique pour passer le temps, en-dehors de leurs habitudes ou pour combler un sentiment de solitude; d'autre part les plus engagés, qui écoutaient plus d'heures par semaines (en moyenne 21 heures, maximum 40 heures), étaient plus susceptibles d'écouter de la musique choisie par eux et d'utiliser la musique pour provoquer des états d'âme spécifiques, créer une atmosphère, ou améliorer telle activité. L'étude a confirmé l'utilité de l'ESM pour examiner les facteurs complexes (et interactifs) qui interviennent dans les choix musicaux quotidiens des individus, et a mis en évidence les façons dont la musique peut réaliser simultanément différentes fonctions. Les résultats montrent que les recherches futures sur le comportement musical de tous les jours devraient se poursuivre en tenant compte du contexte de l'audition musicale dans le façonnage des réponses et des usages de la musique; en tenant compte aussi des différences individuelles dans les niveaux d'engagement des individus à la musique.

Erforschen des alltäglichen Umgangs mit Musik mithilfe der Erlebnisstichproben-Methode

Neuere qualitative Forschung verweist auf Unterschiede im Hörverhalten, die vom Grad der Involviertheit mit und dem Interesse an der Musik zusammenhängen. Diese Ergebnisse basieren jedoch hauptsächlich auf retrospektiven Berichten über Verhaltensmuster (Greasley, 2008; Greasley & Lamont, 2006; Lonie, 2009). Die Erlebnisstichproben-Methode (ESM) ist ein wertvolles Mittel, um Musik in Alltagszusammenhängen zu untersuchen. Dabei zeigte sich, dass durch die ESM auch das Bewusstsein der Menschen über die Rolle von Musik in ihrem Leben wächst (Sloboda, O'Neill, & Ivaldi, 2001). Die vorliegende Studie untersucht unter Verwendung der ESM Unterschiede im alltäglichen Umgang mit Musik, indem drei verschiedene Hörertypen befragt wurden: Menschen, die durch geringen, moderaten oder starken Umgang mit Musik gekennzeichnet waren. Über sieben Tage wurden quantitative und qualitative Daten von 25 jungen Erwachsenen (18-30 Jahre alt) gesammelt. Diese Daten bezogen sich darauf, was sie während des Hörens von Musik taten (z.B. Aktivitäten) und auf die Funktionen/Effekte der Musik (z.B. Gründe, die die Auswahl beeinflussten). Nach Durchführung der Studie führten Interviews mit 23 Teilnehmern zu retrospektiven Berichten über spezifische musikalische Erfahrungen. Die Analysen zeigten zwei weitgefasste Hörtypen: (a) die weniger aktiven Hörer verbrachten weniger Stunden pro Woche mit Musikhören (Minimum: drei Stunden, Mittel: zwölf Stunden), hörten in der Regel weniger selbst gewählte Musik und hörten zum Zeitvertreib, aus Gewohnheit, oder damit sie sich weniger allein fühlten; (b) die hochaktiven Hörer verbrachten eine größere Anzahl von Stunden pro Woche mit Musikhören (Mittel: 21 Stunden, Maximum: 40 Stunden), wählten mit größerer Wahrscheinlichkeit selbst die Musik aus, verwendeten Musik zum Hervorrufen spezifischer Stimmungen, zum Schaffen von Atmosphäre oder zum Verbessern einer Aktivität. Die Studie bestätigt die Nützlichkeit der ESM bei der Untersuchung von komplexen und interagierenden Faktoren, die bei Menschen in ihrem Alltag eine Rolle im Musikauswahlverhalten spielen. Sie legt besonderes Augenmerk auf die Art und Weise, wie Musik gleichzeitig verschiedene Funktionen erfüllen kann. Die Ergebnisse zeigen, dass künftige Forschung zum musikalischen Alltagsverhalten weiterhin den Kontext des Musikhörens berücksichtigen sollte, da kontextbedingt Reaktionen auf die Musik und die Verwendung von Musik beeinflusst wird. Weiterhin sollten individuelle Unterschiede im Ausmaß des Umgangs mit Musik berücksichtigt werden.